



Power Play: Investing in Energy Infrastructure for the AI Era

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The world is facing a significant energy deficit. This was true before AI and the war in Iran exacerbated the shortfall. The kinds of investments underway today will shape the future of the energy landscape and its ability to support an evolving tech-forward economy. Businesses are responding through legacy and innovative ways both to alleviate current constraints and to scale up for the long run.

To address immediate constraint issues, companies are sidestepping grid backlogs by offering readily dispatchable power sources and modular units. As the grid expands to meet growing demand, multi-industry and electrical equipment companies are enabling utilities to expand generation, transmission, and delivery. Renewables cannot be overlooked as a scaling solution as the AI era exponentially increases the demand for energy. Solar in particular is among the fastest-to-deploy sources of new generation. When paired with battery energy storage systems, these technologies can provide the kind of around-the-clock, predictable power that AI infrastructure requires. Commercial nuclear is another important component of the longer-term energy supply makeup, with the near-term focus on extending the lifetime of existing plants while looking ahead to longer-term expansion. All that said, even as the grid scales significantly and the mix of energy supply diversifies, natural gas is likely to remain a primary and backup source of power.

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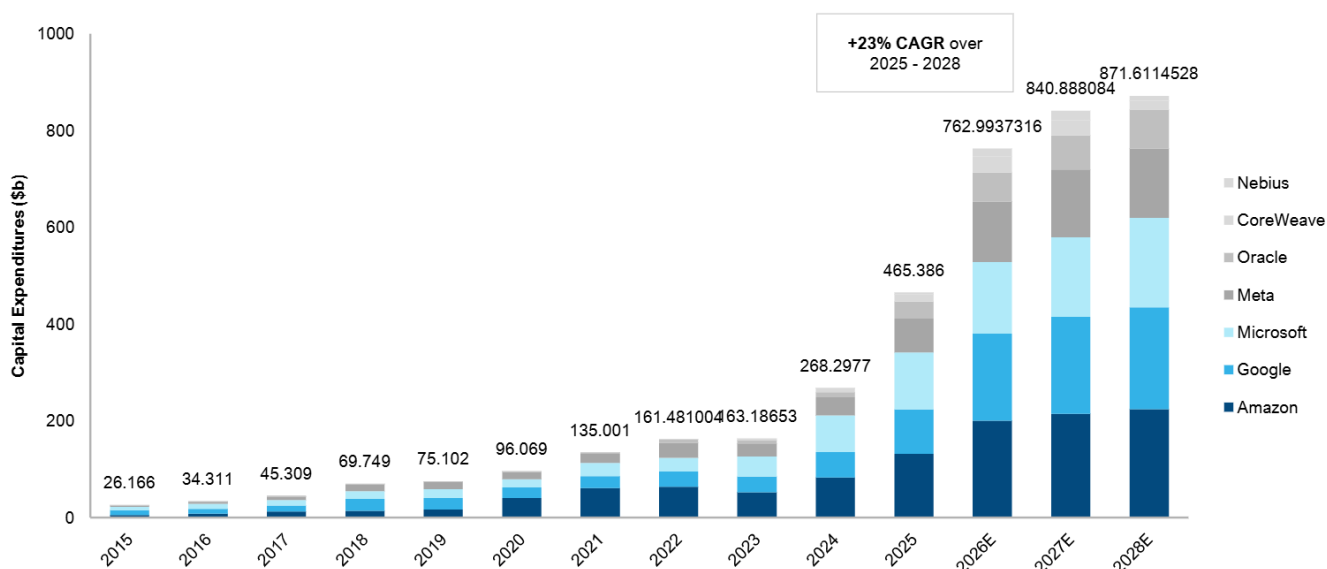
The Macro and Demand Landscape

The grid is under significant pressure. US energy generation hit a record 4.43 terawatt hours in 2025, according to the US Energy Information Administration; generation will likely break that record again this and next year due primarily to large data centers. Energy demand driven by AI data centers globally is expected to quadruple over the next four years, according to the International Energy Agency. To get a sense of how great this demand is and will be in the coming years, we can look to cloud-related investments, growth in data centers and AI workloads, and energy prices.

One of the most telling data points has been cloud-related investments. Our Semiconductors Research Team has tracked a decade of data-center-related investments via capital expenditures (capex) at several major cloud service providers to finance the setup of physical data centers as well as the electronic equipment needed to fill them. The data shows that the current buildout is unlike any other. Capex has surged over the last three years, rising from \$161 billion in 2022 to \$465 billion in 2025. This reflects a compound annual growth rate of +42%, a jump up from comparable three-year rates of +29% (2019-2022) and +30% (2016-2019).

Spending is expected to step up even further to \$762 billion (+64% year-over-year) in 2026. If that happens, this would mark the third year in a row that cloud-related capital investments have risen more than 60% year over year. Prevailing estimates also expect growth to slow after this growth spurt, but as we saw throughout 2025, consensus tends to underestimate the amount of spending in latter years, so future spending could well exceed today's expectations.

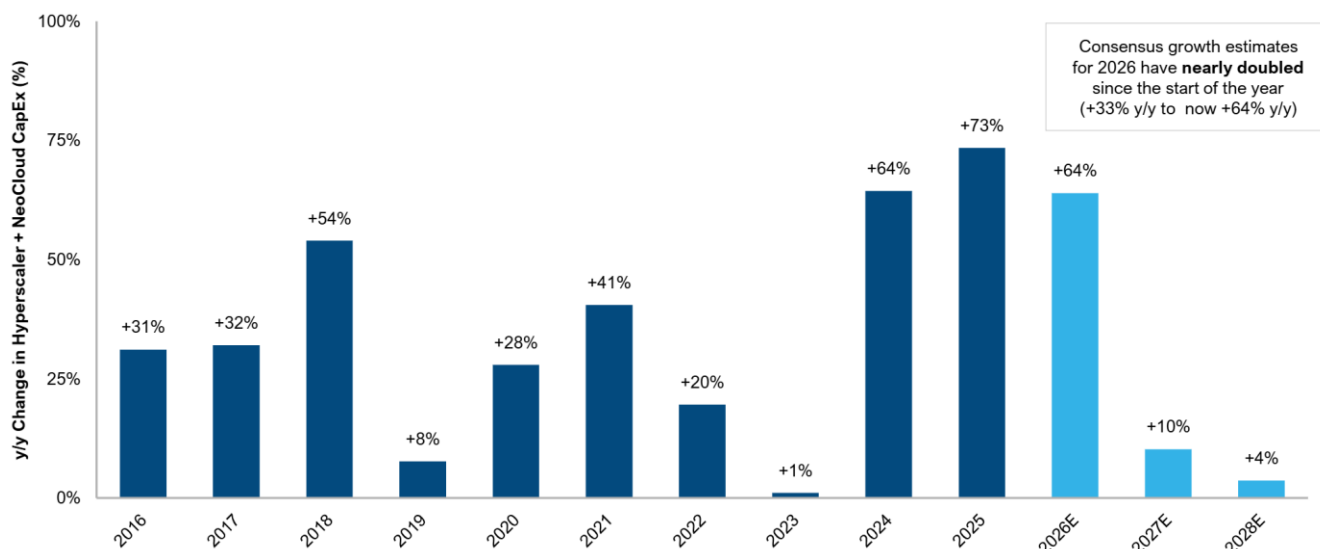
Figure 1: Aggregate Cloud Capex (2015 - 2028E)



Source: Deutsche Bank Research, Bloomberg Finance LP. Note: Dataset available upon request. Forecast data is based on median Bloomberg consensus estimates. Information is presented on an annual basis given the lack of quarterly detail by company and dispersion in consensus estimates going forward.



Figure 2: Annual Growth in Aggregate Cloud Capex (2016 - 2018E)



Source: Deutsche Bank Research, Bloomberg Finance LP. Note: Dataset available upon request. Forecast data above is based on median Bloomberg consensus estimates. Information is presented on an annual basis given the lack of quarterly detail by company and dispersion in consensus estimates going forward.

Data center AI workloads paint a similar AI-driven energy demand picture. Data centers represent about 5% of total US electricity usage today, according to estimates from BloombergNEF and the Department of Energy. This will likely expand with the adoption of AI, as well as with continued growth in the digital economy.

Based on midrange estimates from McKinsey, total global data center capacity demand is set to increase from 82 gigawatts in 2025 to 220 gigawatts in 2030, amounting to a 22% annual growth rate. This estimate sees generative AI workloads growing at 29% on average, plus 11% from non-AI workloads. Generative AI workloads could grow to 35% of total global data center activity by 2028, up from a mid-single-digit percentage as of 2023, according to estimates from Boston Consulting Group.

AI workloads require greater power density (essentially the amount of electricity a system needs per square foot) and more intensive liquid cooling, driving up energy consumption. Data center companies like Digital Realty (DLR) and Equinix (EQIX), for example, are already seeing these trends play out in their businesses. Digital Realty recently said it is rolling out higher-density deployments with pre-installed liquid cooling to meet AI demand. Equinix has noted that 60% of its larger deals in the last quarter of 2025 were driven by AI, up from 50% earlier last year. The power density for these deals, according to Equinix, is 33% higher than for non-AI deals.

These factors exacerbate concerns around US electricity prices and grid capacity, which were already mounting even before the war in Iran elicited a spike in global energy prices. After an extended period of stable growth in the US – over the last decade, electricity output grew at about a 0.7% annualized rate – data center electricity demand estimates are raising concerns about a knock-on



impact to household utility bills as well as grid capacity. The energy problem is just as acute for central bankers. Traditionally, Federal Reserve policymakers would tend to look through the temporary inflationary impact of an energy shock (see US Economic Perspectives: Impact of oil price shocks a "frack"-tion of what they once were). However, the backdrop to the latest energy shock is one in which inflation has persistently exceeded the Federal Reserve's 2% goalpost for the better part of the last five years. This raises the risk that inflation expectations could become unanchored. In short, the energy supply outlook carries implications across a breadth of economic issues, from interest rates to technology stocks.

These concerns could result in policy favorable to companies increasing energy supply, as energy prices will likely be top of mind for voters ahead of the midterm elections this November. As we recently outlined, historically, the relationship between gasoline prices and consumer energy spending suggests that a 1 cent rise in gasoline prices (on a sustained basis) increases spending on energy goods and services. This thus reduces income that can be spent on non-energy items – by \$1.15 billion, amounting to a sizable "energy tax." Mapping the impact of potential energy price shocks to consumers, we find that at a sustained \$100 per barrel oil price, the projected tax benefits to consumers from last year's tax legislation (the "One Big Beautiful Bill Act") would still exceed the drag from the energy tax increase. However, at \$150 per barrel, the implied energy tax increase would more than wipe out the tax cuts to consumers from that legislation and present a more serious threat to the outlook for consumer spending.

The sum total of this is that supply is already struggling to keep up. Wait times to connect a data center to the grid are exceeding four years (based on comments from Google representatives and other industry reports). Efforts are underway to increase supply, including expanding grid capacity, as well as AI players preparing to build their own sources of power that operate "behind-the-meter" (developed on the customer's side and not delivered through the public grid). Capex into investor-owned electric utilities has been steadily increasing and is projected to climb to nearly \$250 billion in 2029 (according to the Edison Electric Institute). The White House's introduction in early March of a Ratepayer Protection Pledge urging AI companies to build or buy additive energy further highlights the race to bolster energy supply to meet AI needs. While the exposure across regions will vary, the conflict in Iran will add another layer of constraint to energy supply, as 20% of global oil and 20% of LNG pass through the Strait of Hormuz. Companies are looking to fill the gaps in energy supply over a spectrum of problem sets solving for immediate constraints, for scale, and for consistency.



Solving for Constraint: Readily Deployable Power

The most direct response to AI-driven power constraints is coming from companies that enable data center operators to bring electricity online without waiting for grid expansion. These companies are delivering dispatchable on-site or near-site power. While there are existing manufacturers providing these kinds of products, there are an increasing number of companies adapting existing technology to enter this AI-adjacent space. FTAI Aviation (FTAI), which traditionally operates in the airline industry, and BorgWarner (BWA), an auto industry supplier, are examples of this kind of shift.

FTAI Aviation launched FTAI Power in December 2025 with the aim of converting the company's CFM56 engines into gas turbines that can deliver energy to data centers. The engine is prevalent, with enough in operation globally that one takes off every two seconds somewhere in the world. FTAI Power will offer its aeroderivative turbine "FTAI Mod-1," a mobile, trailer-mounted conversion unit that can be delivered and installed in under two weeks. As a 25-megawatt unit, it is smaller, which allows for flexibility in stacking and growing data centers. Deliveries will likely begin around December 2026, with approximately 100 units in production per year starting in 2027.

BorgWarner is taking a similar approach. The company is an auto industry supplier that designs and manufactures powertrain systems but will partner with Turbocell (a subsidiary of the AI-optimized data center infrastructure company Endeavour) to co-develop an integrated, modular power generation system. This system would provide primary and backup power for AI data centers and would serve as a "bridge" solution, meaning data center operators can install it quickly for power now but can keep it on to provide additional power even as longer-term energy infrastructure is built up.

Modular power generation systems like this one are designed with the high and often fluctuating power demands of AI computing in mind. Modular products also comprise standardized, interchangeable units that are more flexible in applications than the custom builds commonly found in legacy systems. These modular units, as well as the integrated battery systems and electronic components, can be scaled up or down based on specific data center needs.

Beyond quick-install energy solutions, companies further up the supply chain are adapting to the energy demanded by growth in AI. Woodward, for example, is an aerospace-industrial hybrid company whose power generation products have historically been focused on the industrial gas turbine market, but it is now broadening into supplying components for reciprocating engines and marine engines being repurposed as behind-the-meter power solutions. |



Solving for Scale: Grid Expansion and Infrastructure Development

Meeting AI-driven electricity demand forecasts will require system-wide capacity and infrastructure expansion. Investments are happening simultaneously but across pathways that range from near-term grid expansion and enablement platforms, near- and longer-term renewables, and long-duration commercial nuclear supply. In terms of grid expansion, efforts span both generation capacity and transmission and distribution (T&D) infrastructure to accommodate higher loads and load density, more stringent reliability requirements, increasingly complex grid architectures, and longer-duration power needs associated with AI workloads. GE Vernova (GEV), for example, manufactures combined-cycle and peaker gas turbines (currently the most common source of baseload power in the US) and also provides high-voltage T&D equipment including transformers, high-voltage direct current (HVDC) solutions, circuit breakers, switchgear, power-sending devices, and grid automation systems. Some of its products also supply the renewable and commercial nuclear sources of power.

As existing grid infrastructure faces a critical deficit, it is impossible to ignore renewables as a scaling solution. Although US onshore wind investment faces challenges due to unfavorable policy headwinds, solar, alongside other renewables, is among the fastest-to-deploy sources of new generation. When paired with battery energy storage systems (BESS), these technologies can meet the 24/7 power needs of AI-focused data centers. There are indications that as energy demand driven by AI increases, so too do investments in renewable energy sources. A recent study from the University of Zurich, for example, found that solar and BESS have been the primary drivers of increased energy investment in regions with pre-existing data center capacity (based on ChatGPT infrastructure in the US). Increased investments in gas play a role too, but these have been offset by declines in grid-connected gas plants. These developments are currently based on investment decisions, but policy drivers could add further impetus should the environmental toll stemming from AI usage take an increasingly prominent place in political discourse.

Solar energy is well suited to the energy demand landscape due to the technology's fast scalability, low to zero marginal costs, and favorable levelized cost of energy (LCOE). As a result, the sector is seeing an increasing number of power purchase agreements (PPAs) with large technology players, including Amazon (AMZN) and Google (GOOGL). These arrangements are also conducive to a buildout of development projects that are supporting equipment suppliers.

Clearway Energy (CWEN), for example, operates energy generation assets in the US, with a specific focus on clean tech. It is positioning itself to serve the growing demand for power in the US driven by hyperscalers and data centers. The company has announced 2 gigawatts of PPAs for these projects, with possibilities for additional volumes. Additionally, it has outlined 8-13 gigawatts of capacity via co-located data center complexes, utilizing a mix of renewables and storage in addition to gas.

The scale-up of renewables also depends on component and equipment suppliers. First Solar (FSLR), for example, is a US-based solar technology



company that designs and manufactures advanced thin-film modules (a kind of solar photovoltaic panel) for utility-scale projects. Nextpower (NXT), meanwhile, is an example of a clean tech equipment and systems provider, selling trackers (the mechanical systems that hold and rotate panels to follow the sun), software, electrical systems, and the foundations upon which solar technology is mounted. Solar electrical equipment also includes inverters, which convert solar-generated direct current electricity into distributable alternating current electricity delivered to both residential and commercial customers.

Commercial nuclear is another important component of longer-term energy supply, with small modular reactors (SMRs) a key part of the development pipeline. Businesses are focused on extending the lifespan of current reactors while also preparing for the possible buildout of new ones. In the near term, the priority is the maintenance, repair, and life extension of existing nuclear infrastructure. There are 54 nuclear power plants in the United States (amounting to 94 nuclear reactors), according to 2024 data from the US Energy Information Administration. These produce a net generating capacity of nearly 97 gigawatts and make up about 20% of US energy usage.

Maintenance of these plants requires the replacement and upgrading of critical reactor components, modernization of control systems, and other projects that allow current reactors to operate safely and consistently for longer periods. Suppliers such as BWX Technologies (BWXT) and Curtiss-Wright (CW) provide operating components and systems for nuclear plants, including reactor hardware, fuel-related technologies, pumps, valves, and control systems.

At the same time, the industry is preparing for potential new build activity, particularly around advanced reactor designs and SMRs. Development work spans reactor components, fuel technologies, and digital and mechanical systems designed for newer reactor platforms. BWX Technologies' work on advanced fuel and reactor components and Curtiss-Wright's involvement in systems and controls applicable to newer reactor designs are examples of how suppliers are positioning for a more modular and scalable nuclear build-out over time.



Solving for Consistency

At the core of scaling AI infrastructure is the need for stable, reliable, and low-cost power. While wind and solar power generation will undoubtedly be part of the mix, natural gas can be expected to continue to play a sizable role as both a primary source of power and a backup source of power to intermittent power sources such as wind. Even with significant renewable penetration, the grid will still require backup generation capacity. Natural gas, in our view, will remain the likely fuel selected for this purpose, due to its global availability (via pipeline or liquefied natural gas imports), energy density, price, and ability to be transported and stored on site for behind-the-grid power. Natural gas also avoids many of the political hurdles that constrain other energy sources. It does not face the regulatory and political hurdles of traditional nuclear and SMRs, nor is it falling out of favor in Western markets to the same degree as coal. As such, it is likely that natural gas will continue to be in demand for both baseload and backup power generation.

The Bottom Line

Projected energy demand tied to the growing AI ecosystem and the broader digital economy is rising faster than grid capacity and interconnection timelines can adjust. Businesses are responding to address immediate constraint issues, questions of scale, and the need for consistent primary and backup generation. This will shape not only the trajectory of the AI buildout, but have reverberating impacts across the broader economy, its competitiveness, and critical infrastructure.



Appendix 1

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What Iran means for the dollar: a perfect storm for the petrodollar

March 24, 2026

The long-term legacy of the Iran conflict for the dollar could be the way it tests the foundations of the petrodollar regime. If fault lines are further exposed, there could be significant downstream effects to the dollar's use in global trade and savings, and the dollar's role as the world's reserve currency.

The world saves in dollars in large part because it pays in dollars. The dollar's dominance in cross-border trade is arguably built on the petrodollar: globally traded oil is priced and invoiced in USD. This arrangement can be traced to a deal struck in 1974 where Saudi Arabia agreed to price oil in USD and invest surpluses in USD assets, in exchange for US security guarantees. Because oil is a core input to global manufacturing and transport, there is a natural incentive for global value chains to dollarize, and global surpluses to accumulate in USD.

The foundations of the petrodollar regime have been under pressure even before this conflict. Most Middle East oil is now sold to Asia not the US; sanctioned oil from Russia and Iran has already been trading off dollar rails; Saudi Arabia has been localizing defence, and experimenting with forms of non-dollar payment infrastructure such as Project mBridge.

The current conflict may expose further fault lines, by challenging the US security umbrella for Gulf infrastructure and the maritime security for global trade in oil. Damage to Gulf economies could encourage an unwind in their foreign asset savings held largely in dollars. In this context, reports that the passage for ships through the Strait of Hormuz may be granted in exchange for oil payments in yuan should be closely followed. The conflict could be remembered as a key catalyst for erosion in petrodollar dominance, and the beginnings of the petroyuan.

A bigger risk could come if the world begins to move away from globally traded oil and gas itself, to more resilient sources of energy including domestically available fuels, renewable energy, and nuclear power. The energy choices of the Global South, Europe and North Asia will be key to track. A move away from oil could be as powerful as the pressure to price it in other currencies.

A world that becomes more self-sufficient in defence and energy would also be a world that holds less USD reserves. The huge strategic importance of the Middle East to the dollar's role as the world's reserve currency should not be underestimated. The current conflict may be the perfect storm for the petrodollar.

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What Iran means for the dollar: a perfect storm for the petrodollar

The dollar is the world's reserve currency for a very simple reason: the world pays for global goods and services in dollars and is willing to save resulting surpluses in dollar assets. From 1945-1971, the dollar was "backed" by gold – namely, global central banks were able to exchange \$35 for 1 oz at the Fed. This was the foundation of the international monetary system known as Bretton Woods. In 1971, the US broke the dollar's link to gold. Since then, the dollar has been in a purely fiat regime – one that is backed by the sovereign credit worthiness of the US and willingness of the world to save in its debt.

Enduring support for the fiat dollar arguably comes from the dominance of the dollar in the pricing of cross-border trade. Globally traded goods and services are largely priced in USD, with payments exchanged over US controlled payment rails. Global surpluses are thus built in USD and mostly invested back into US assets. Corporates are incentivized to save and borrow in the currency of their payables and receivables, banking systems are dollarized, and central banks save in dollars to act as effective lenders of last resort. This drives demand for USD reserves. See more on this link [here](#).

A crucial anchor to this system is the petrodollar: the fact that most globally traded oil is priced and invoiced in dollars. Because oil is so central to global manufacturing processes – from petrochemicals, fertilizers and transport, to running factories and offices – companies are incentivized to price end products in dollars as a natural currency hedge to a key cost. This is a big reason global goods and services are priced and traded in US dollars.

The reason oil is priced in dollars can be traced back to the 1974 petrodollar arrangement between the US and Saudi Arabia. In simple terms, Saudi Arabia agreed to price its oil exports in USD and invest oil surpluses into US Treasuries. In exchange, the US provided security guarantees and military protection. The rest of the GCC followed. Structural foreign demand for US debt lowered the funding costs of the US government, acting as an indirect means of payment for the security umbrella extended to the region. To the extent that this arrangement supported broader global invoicing and saving in USD, it has been crucial to the USD's role as the world's reserve currency.

The current conflict has arguably shaken some core foundations of the petrodollar regime: the security-for-oil-pricing arrangement. US military assets and bases in the Gulf have come under attack in the war. Oil infrastructure in the Gulf has also been hit. And the US ability to provide the maritime security to ensure the global flow of oil has been challenged with the closure of Hormuz. The US security umbrella has been fundamentally tested.

The legacy of this conflict for the dollar could be the ways in which it tests the foundations of the petrodollar regime. In the long-run, if the world uses less oil, the Gulf draws more deeply on existing dollar savings, if the Gulf moves closer to Asia in its trade and investment relationships, and eventually prices less oil in dollars – there could be significant downstream effects to the dollar's usage in global trade and savings.

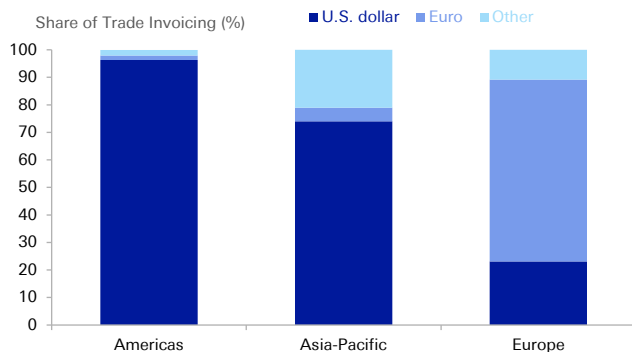


The system was already under threat: many changes have been afoot

A lot had already begun to change before the US-Israel-Iran war broke out. We highlight a few key shifts that were underway:

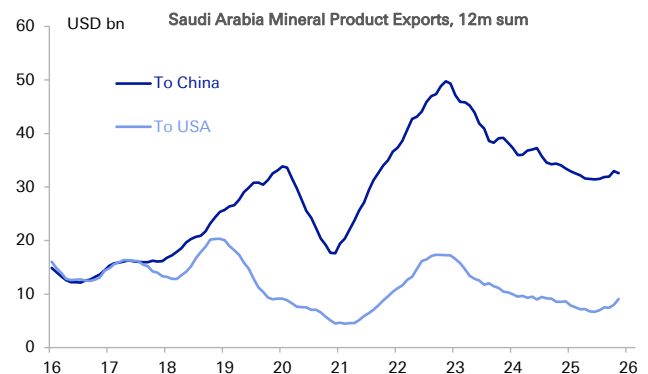
1. **The US was no longer the biggest buyer of Middle East oil.** With the shale revolution making the US energy independent, Saudi Arabia had been selling more than four times as much oil to China as to US. 85% of Middle East crude oil goes to Asia. This already introduced a fundamental instability with reports that China was keen for more oil to be invoiced in yuan.
2. **Saudi Arabia was already looking to localize more of its own defence.** Under Vision 2030, Saudi Arabia had been targeting an increase in the domestic content of military spending to 50%, looking to reduce dependence on foreign imported arms.
3. **Saudi Arabia had joined Project mBridge and signed FX swap lines with China.** Project mBridge is an initiative involving the PBOC, HKMA, Bank of Thailand, and the central banks of the UAE and Saudi Arabia. It uses blockchain technology to facilitate payments in the central bank digital currencies of each country. Crucially it does not depend on USD correspondent banking or SWIFT, and is at the minimum viable stage. The rails to transact outside of the dollar have already been built.
4. **Sanctions on Russia and Iran meant significant oil trade was already taking place outside of dollar rails.** Sales of Russian and Iranian oil have been priced and transacted in a range of local currencies from ruble, yuan, rupee with non-dollar payment infrastructure having been used.

Figure 1: The dominance of the dollar in cross-border trade has a lot to do with the petrodollar



Source : Deutsche Bank, Federal Reserve

Figure 2: There were already signs of instability to petrodollar foundations: Saudi Arabia sells four times as much oil to China now as to the US



Source : Deutsche Bank, CEIC



The conflict has introduced new sources of instability for the petrodollar

First, the US security umbrella is being tested with Gulf economies facing attacks on US bases, oil fields and infrastructure in a war that began with US military action in the region. Aside from the Gulf, pain has arguably been disproportionately felt by traditional US allies: Europe, Japan and Korea that are large energy importers exposed to the Straits.

Second, direct diplomacy rather than US-led maritime security has been the conduit for oil to pass through the Strait of Hormuz. Select tankers heading for China, India and Japan have been allowed passage. Bilateral relations have served countries.

Third, and crucially there have been reports that Iran is negotiating with eight countries to allow passage of ships through the Strait of Hormuz if oil payments are paid in yuan ([The Telegraph](#), [CNN](#), [Asia Times](#), [Chosun](#)). This adds further credence to the enormous power of oil pricing. The conflict could be the catalyst for erosion in petrodollar dominance and the beginnings of the petroyuan.

To be sure, there are numerous mitigants to consider. While the US is no longer the biggest global buyer of oil, it may well be on track to be the world's biggest supplier. As we argued in our [earlier piece](#), if the US controlled the entire Western Hemisphere's oil – either directly or by proxy – it would have more reserves than OPEC, putting it in a better position to set the terms of global oil trade.

In a best-case scenario, the dollar could remain the dominant currency for oil trade if the US controls a majority of the world's supply, although further development of reserves would be required to produce surpluses sufficient to be the world's biggest *tradeable* supplier.

In a worst case scenario, one may see a fracturing of global oil pricing along trade routes and corridors. One might imagine Middle East oil passing through the Straits of Hormuz to Asia being priced in yuan, while oil from the Western Hemisphere sold across the Atlantic and Pacific to historical US allies may be priced in USD.

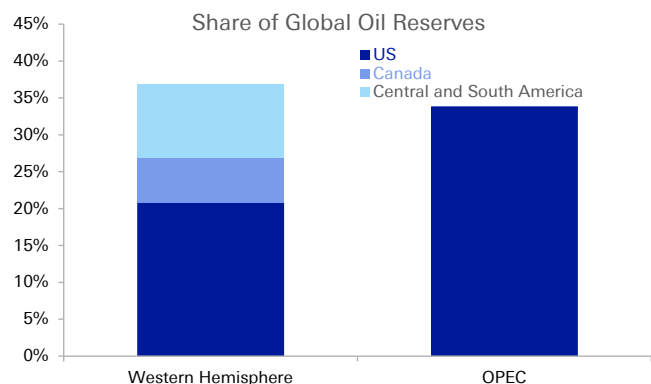
Another important mitigant is that Gulf economies are very invested in the USD. The GCC runs USD pegs that are backed by enormous USD savings. To maintain pegs, it makes sense for them to add weight with USD receivables. Any sign of moving away from USD revenues could invite self-fulfilling attacks on pegs.

However, the Gulf may need to unwind dollar savings due to this crisis itself. The war has already led to damage of oil and civil infrastructure in Gulf economies. Escalation or a prolonged conflict could worsen this (see [here](#) for what's at stake for Gulf economies and [here](#) for our take on escalation risks). Hydrocarbon economies could suffer both from near-term hits to physical production and reduced long-term demand for fossil fuels (more below). Moreover, services sectors such as finance, tourism, aviation may face scarring if there remains an overhang of security risk. All of this could mean Gulf economies need to redeploy more of their externally held dollar savings domestically to support local economies and potentially reorient diversification programs to a different regional climate. To be sure, reserves are plentiful: the MENA region has roughly USD2tn in central bank managed reserves, and around USD6tn in sovereign wealth funds, according to Global SWF. But if USD



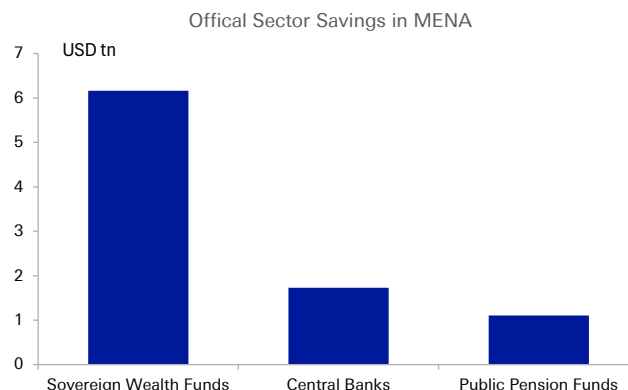
savings are being drawn down for domestic investments, this could reduce the hurdle to other currency changes.

Figure 3: If the US controlled Western Hemispheric oil, it would have more reserves than all of OPEC



Source : Deutsche Bank, Energy Institute

Figure 4: A majority of the Gulf region's savings are held in sovereign wealth funds



Source : Deutsche Bank, Global SWF

Perhaps the biggest long-term risk comes not just from the Gulf moving away from dollar pricing, but if the world begins to move away from oil itself to which we turn next.

Bigger energy shifts could be in the offing: the world may diversify away from traded fossil fuels to domestic energy

There are many parallels between now and the 1970s. Most obviously: we have now seen a second major oil and gas shock this decade. In the 1970s, the 1973 oil embargo was followed by the 1979 Iranian Revolution, while this time, Russia's invasion of Ukraine (2022) has been followed by the ongoing US-Israel-Iran war. Global prices may remain structurally higher beyond the end of the current conflict if production facilities in the Gulf are materially damaged. And the historic weaponization of the Straits could introduce more risk premium into energy that moves on ships. Increasing self-sufficiency and domestic resilience would make sense even if oil prices came down.

The 1973 oil embargo by Arab nations on Western economies motivated big energy shifts in efficiency, diversification, and reserve building. It accelerated the development of alternative oil sources in Canada, Gulf of Mexico, Alaska, and the North Sea that helped the OECD diversify away from Middle East oil. It also led to the creation of Strategic Petroleum Reserves, and created the political will for initial investments in renewable and nuclear power.

The US may see the least change. Unlike the 1970s, the US is now energy independent in oil & gas. US industry will not be hurt immediately in the way that economies not receiving physical oil will be, and US oil producers will benefit. While there could be a consumption shock from higher pump prices, the US arguably retains the option of an export ban to manage domestic prices. WTI-Brent-Oman spreads already reflect the very different prices being paid around the world. The US growth impact is thus the most ambiguous and likely the least negative (see [here](#) and [here](#)). This should mean the US remains relatively more invested in fossil fuels.



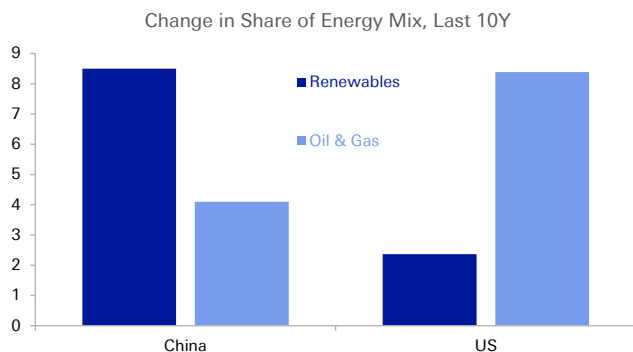
But energy-dependent regions like Europe, Asia and many parts of the Global South – that are facing multiple threats to global trade in fossil fuels – arguably have three key options:

- **The first option is to source more fossil fuels domestically.** Regions that have the ability to explore and develop more of their own oil and gas may be more motivated to do so, from the UK to Brazil, while Europe and parts of Asia may also turn back to domestic sources of coal. Even if the world does not move away from fossil fuels, global trade in it could well decline, and this is what matters for FX.
- **The second option is to double down on renewable energy.** Renewables are far more appealing on relative cost today than they were in the 1970s, due to the enormous industrial build out in China. China produces 80% of world's solar panels, 70% of the wind turbines, and 70% of lithium batteries ([Science](#)). China's overcapacity may well have been strategic, with China now extremely well placed to supply into new demand. Many Global South economies could accelerate this shift. But ultimately, this second option also comes with new dependencies on a mostly single industrial actor.
- **Genuine energy independence may only be created by a third option: nuclear power.** One could perhaps expect a significant focus on this in Europe, Japan (see [here](#)), and South Korea – traditional US allies. Just like their defence buildouts, nuclear power will have very long multi-year lead times but could have very significant FX implications.

To be sure, there will be limits to how much the world can fully move away from oil. Many industries rely on the fractionalization of oil for manufacturing. And as we have argued in our [earlier work](#), oil is crucial to militaries: while passenger cars can run on electricity - military jets, tanks and navy ships are likely to be powered by oil for a long time. Nevertheless, there is scope for greater parts of the energy mix to transition to more reliable, secure, and potentially cheaper sources of energy. If the world shifts away from globally traded oil and gas, to more domestic sources of fuel, renewable or nuclear power, the most obvious long-term impact would be reduced oil and gas deficits in Europe and North Asia, and reduced energy surpluses in the Middle East. Reduced global oil trade would also create more room for pricing of goods and services to shift away from the USD.

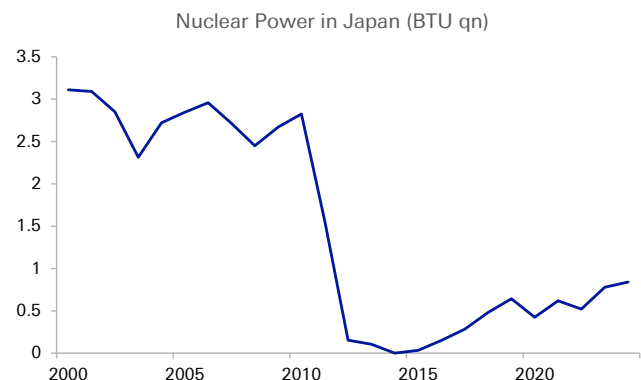
The move away from oil could be as powerful as the pressure to price it in other currencies. The petrodollar system could be at risk from both the 'petro' and the 'dollar' legs.

Figure 5: China has been leaning more towards renewable energy, while the US has been doubling down on fossil fuels



Source : Deutsche Bank, CEIC, IEA

Figure 6: Japan and Europe could ramp back up on nuclear power to build more energy independence



Source : Deutsche Bank, CEIC



Conclusion: What does this all mean for the dollar?

In the near term, we have [argued](#) that US energy independence might be worth some safe-haven premium on the dollar. The US is the only major economic region in the world that is both energy independent and is itself far from a battleground. However, there have been important offsetting factors from growing US fiscal risk around greater military spending, to the unwind of UST holdings in Asia and the Middle East to defend currencies (see pieces [here](#) and [here](#)) which mean the dollar has not strengthened much in this crisis even in the short term.

The more interesting and lasting impact of this crisis on the dollar could be on the long-term foundations, namely the willingness of the world to price traded goods and services in dollars and to save resulting surpluses in dollar assets.

The foundations of the petrodollar regime were under pressure even before this conflict. There were already signs of instability to the long-standing arrangement to price GCC oil in dollars in exchange for security: most Middle East oil is sold to Asia now not the US; sanctioned oil has already been trading off dollar rails; Saudi Arabia has been localizing defence and experimenting with non-dollar payment rails alongside other Global South central banks.

The current conflict may have exposed further fault lines, by challenging the US security umbrella for Gulf infrastructure, the maritime security for global trade in oil, and encouraging a potential unwind in Gulf dollar savings. In this context, reports that the passage of ships through the Strait of Hormuz may be granted in exchange for oil payments in yuan should be closely followed. The conflict could be the catalyst for erosion in petrodollar dominance and the beginnings of the petroyuan.

A bigger risk perhaps comes if the world begins to move away from globally traded oil and gas itself, to more resilient sources of energy from domestically available fuels, renewable energy, and nuclear power. The energy choices of the Global South, Europe and North Asia will be particularly crucial to track. The move away from oil could be as powerful as the pressure to price it in other currencies.

A world that is looking to become more self-sufficient in defence and energy would also likely be a world that will hold less USD reserves. If economic security is now a key part of national security, building greater energy resilience should be seen as complementary to the increased focus on defence spending. Reserves hitherto held in foreign assets will increasingly be needed to build domestic capacity that supports strategic autonomy (see [here](#) and [here](#)). Future surpluses could themselves reduce and be accumulated less in dollars. These are long-term dollar negatives.

The huge strategic importance of the Middle East to the dollar's role as the world's reserve currency should not be underestimated. The current conflict may be the perfect storm for the petrodollar.



Thematic FX Framework Pieces

This note is part of a broader series of thematic FX notes that explores the evolving nexus between payments, energy and defence as key pillars to the dollar's long-term status. Other FX pieces that can be read alongside this include:

[FX Special Report: What Venezuela has to do with the dollar](#)

[FX Special Report: What do stablecoins mean for dollar dominance?](#)

[FX Special Report: Who can come home?](#)

[FX Special Report: The geopolitics of reserves: shifting security alliances will matter for the USD](#)

[EM Special Publication: Geoeconomics 101: implications for the dollar, emerging markets and beyond](#)

[FX Blog: On defense spending, bond markets, home bias & the yen](#)

[EM Blog: Iran - a further impetus away from recycling savings to the US?](#)



Appendix 1

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